Problem Set 4

Due Online: Thursday, March 27, 11:59 pm

1. Show that a GARCH(1,1) model can be written as an infinite ARCH process (assuming the weak stationarity condition holds). *Comment*: This explains why GARCH models are typically more empirically relevant than ARCH models.

2. Consider the model

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \\ \varepsilon_t &= \sqrt{h_t} v_t, \quad v_t \sim IID\,(0,1)\,, \\ h_t &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2, \end{aligned}$$

   where $|\phi_1| < 1$, $\alpha_0 > 0$, $\alpha_i \geq 0$, for $i = 1, 2$, and $\alpha_1 + \alpha_2 < 1$.

   (a) Derive the conditional mean and variance of y_t given $\Omega_{t-1} = \{y_{t-1}, y_{t-2}, \ldots\}$.

   (b) Derive the unconditional mean and variance of y_t.

3. Consider the GARCH(1,2) process $\varepsilon_t = \sqrt{h_t} v_t$, where $v_t \sim IID(0,1)$,

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \beta_1 h_{t-1},$$

   $\alpha_0 > 0$, $\alpha_i \geq 0$, for $i = 1, 2$, $\beta_1 \geq 0$, and $\alpha_1 + \alpha_2 + \beta_1 < 1$.

   (a) Show that $E\left(\varepsilon_t^2 | \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots\right) = h_t$.

   (b) Derive the unconditional variance of ε_t.

   (c) Derive the h-step ahead forecasts of h_{t+h} at the forecast origin t, for $h \geq 1$.

4. Consider the model for $\mathbf{y}_t = (y_{1t}, y_{2t})'$:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}, \tag{1}$$

   where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t})' \sim WN\,(\mathbf{0}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1.0 & 0.7 \\ 0.7 & 1.5 \end{pmatrix}.$$

   (a) Is the bivariate series $\mathbf{y}_t$ weakly stationary?

(b) Find the mean vector of $\mathbf{y}_t$.

Hint: The inverse of a 2×2 invertible matrix $\mathbf{A}$ has the formula:

$$\text{If } \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } \mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(c) Suppose the sample size is $T = 100$, and in the last period we observed $y_{1,100} = 1$ and $y_{2,100} = 0.5$. What are the forecasts of $y_{1,101}$ and $y_{2,101}$ and their associated standard errors?

(d) Find the coefficient matrices on $\boldsymbol{\varepsilon}_t$, $\boldsymbol{\varepsilon}_{t-1}$ and $\boldsymbol{\varepsilon}_{t-2}$ in the MA representation of $\mathbf{y}_t$.

5. (Computer Exercise) Retrieve the daily stock prices of JPMorgan Chase & Co. (JPM) over the period of 2016–2019 from Yahoo! Finance using the `quantmod` package. Compute the daily log returns from closing prices as differences of log prices.

(a) Plot the return and squared return time series. Do you observe volatility clustering?

(b) Do you find serial correlations and ARCH effects in the returns according to the ACF, PACF, and portmanteau tests? Explain.

(c) Fit a Gaussian ARMA(0,0)-GARCH(1,1) model to the return series. Perform model checking. In particular, check the normal QQ-plot of the standardized residuals. Explain whether the model is adequate.

(d) Fit an ARMA(0,0)-GARCH(1,1) model with standardized Student-t innovations to the returns. Perform model checking, including showing the QQ-plot and the series with 2 conditional standard deviations superimposed. Explain whether the model is adequate. Write down the fitted model.

(e) Compute the 1- to 10-step ahead forecasts of the returns and conditional standard deviation based on the model you obtained in part (d).

Comments: For parts (c) and (d), you do not need to include all figures from your model checking steps. Instead, please explain what you examined and how you reached your conclusion. You may disregard the Nyblom stability test and Sign Bias Test from the `ugarchfit` output.